CryptoAssist: Industry Statistical Assistant

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1 Introduction

- **Industry:** Finance / Cryptocurrency
- **Intended User:** Retail Crypto Investor or Portfolio Manager
- **Problem:** Investors often struggle to understand underlying market risks and trends, leading to poor portfolio decisions based on emotion rather than data.

- **Decision Supported:** Determine which crypto assets to hold, buy, or sell based on volatility, historical trends, and risk assessments.
- **Project Goal:** To build a robust statistical assistant capable of providing objective data analysis and API endpoints to power a cryptocurrency dashboard.
- **Selected Implementation Level:** Level 3 (Full Local Statistical Application with a Next.js frontend and R Plumber backend).

2 Dataset

- **Source:** Synthetically generated cryptocurrency dataset covering 1 year (2025).
- **Unit of Observation:** One row represents the daily trading data (Open, High, Low, Close, Volume, MarketCap, DailyReturn) for one specific cryptocurrency.
- **Number of Observations:** 3 coins * 365 days = 1095 observations.
- **Main Variables:** Date, Coin, Close, Volume, DailyReturn.
- **Data Preparation:** Data is loaded via `R/clean_data.R`. Dates are coerced into Date objects.
- **Missing-data issues:** A small percentage of volume data is set to NA for testing the API's data-quality validation handlers.
- **Data-quality issues:** The dataset is synthetically generated, so it is cleaner than raw API data, but validation rules are in place to handle missing variables.

3 Explore Skill

- **User question:** What is the distribution of daily returns for BTC?
- **Method:** Descriptive statistics and Histogram creation.
- **Descriptive statistics:** Calculates Mean, SD, Min, Max, and counts missing values.
- **Visualization:** Returns histogram breaks and counts for frontend rendering.
- **Validation:** Checks if the coin exists, if the variable is numeric, and fails if more than 50% of the data is missing.
- **Main result:** BTC exhibits a mean daily return near 0.1% with a 3% standard deviation.
- **Interpretation:** The average DailyReturn for BTC is slightly positive, with high volatility indicative of crypto markets.
- **Limitation:** These statistics summarize past behavior and do not predict future returns.

4 Compare Skill

- **User question:** Does BTC have a significantly different average daily return compared to ETH?
- **Variables:** `coin1` (BTC), `coin2` (ETH), `outcome_variable` (DailyReturn).

- **Method:** Welch Two Sample t-test (assumes unequal variances).
- **Group results:** Returns the mean for group 1, mean for group 2, and sample sizes.
- **Estimated difference:** Calculates the exact difference in means.
- **Confidence interval:** Returns the 95% CI.
- **P-value:** Extracts the p-value from the t-test.
- **Validation:** Ensures exactly two distinct valid groups are requested with at least 30 observations each.
- **Interpretation:** If $p < 0.05$, we reject the null hypothesis, indicating a statistically significant difference in performance.
- **Limitation:** This test identifies an association in historical returns, not a causal relationship.

5 Model Skill

- **User question:** What factors are associated with the daily return?
- **Outcome:** `DailyReturn`
- **Predictors:** `Volume`, `MarketCap`
- **Model type:** Multiple Linear Regression
- **Evaluation metric:** R-squared and Adjusted R-squared.
- **Evaluation result:** The API returns the R-squared value along with estimated coefficients.
- **Validation:** Validates that the outcome and predictors exist and are strictly numeric.
- **Interpretation:** The model explains X% of the variance in the outcome using the selected predictors.
- **Limitation:** Linear regression may not capture the highly non-linear dynamics of cryptocurrency markets.

6 Additional Skill 1: Trend Analysis

- **Skill name:** Trend Analysis
- **User question:** Is the current price trend bullish or bearish?
- **Method:** 30-Day Simple Moving Average (SMA).
- **Validation:** Requires a valid date variable and at least 30 days of historical data for the chosen coin.
- **Result:** Returns the current price, the 30-day MA, and a "Bullish" or "Bearish" trend label.
- **Interpretation:** If the current price is above the 30-day moving average, it indicates positive short-term momentum.
- **Limitation:** Moving averages are lagging indicators and cannot accurately predict sudden market reversals.

7 Additional Skill 2: Outlier Detection

- **Skill name:** Outlier Detection
- **User question:** Which days experienced unusually extreme market movements?
- **Method:** Z-Score Outlier Detection (Flagging $|Z| > 3$).
- **Validation:** Ensures the target variable is numeric.
- **Result:** A count of total outliers and a list detailing the date, raw value, and Z-score for each.
- **Interpretation:** Extreme Z-scores indicate market anomalies, crashes, or spikes.
- **Limitation:** Z-scores assume a normal distribution, whereas crypto returns often feature "fat tails."

8 Additional Skill 3: Asset Ranking

- **Skill name:** Asset Ranking
- **User question:** Which coin has the highest risk-adjusted historical return?
- **Method:** Risk-Adjusted Return (Simplified Sharpe Ratio = Mean Return / Volatility).
- **Validation:** Requires data spanning the last 90 days.
- **Result:** A descending ranked list of assets based on their Sharpe ratio over the last 90 days.
- **Interpretation:** A higher Sharpe ratio indicates better historical returns relative to the risk experienced.
- **Limitation:** The simplified Sharpe ratio assumes a risk-free rate of zero and normal distribution of returns.

9 Google Antigravity and Agent Skills

- **How Antigravity was used:** Antigravity was used to generate the Plumber API architecture, write the R statistical functions, and scaffold the Next.js frontend plan.
- **How the skills guided the project:** The 6 `SKILL.md` files acted as rigid prompts, ensuring the API routes returned exactly the data structures needed for interpretation and validation.
- **One useful agent contribution:** The agent correctly suggested a Welch t-test for the compare skill, noting that cryptocurrency variances are rarely equal.
- **One suggestion corrected:** I had to adjust the trend analysis script to ensure the data was sorted chronologically before calculating the moving average.
- **Verification:** I manually tested the Plumber endpoints using curl/Postman to verify the JSON output matched expectations.

10 Application or Delivery

- **Access:** The intended user accesses the assistant via a **Local Next.js Interface** (Level 3 Implementation) running on `http://localhost:3000`. This frontend communicates with an **R Plumber API** running locally on port 8000.
- **Quarto Report:** This document serves as the formal statistical documentation as required by the rubric.

11 Limitations

1. **Dataset Limitation:** The dataset relies on simulated data, which does not perfectly mimic true market microstructures.
2. **Statistical Limitation:** Linear models often fail to capture the heteroskedasticity and volatility clustering inherent in crypto data.
3. **Implementation Limitation:** The system is purely local and requires the user to start both the R backend and Node frontend manually.

12 Reflection

- **What you learned:** I learned how to decouple statistical logic in R from a modern React frontend using an API layer (Plumber).
- **What was most difficult:** Handling JSON serialization edge cases between R lists and Javascript objects.
- **Skill selection:** I selected Trend Analysis, Outliers, and Ranking because they map directly to standard financial indicators used in crypto trading platforms.
- **Improvements:** With more time, I would replace the synthetic dataset with a live data pipeline using the `quantmod` or `crypto2` R packages.